

Bounded Arithmetic, Constant-depth Frege Proofs, and the Linear Time Hierarchy

COMP532 – Propositional Proof Complexity

Nathan Acheampong and Zhaoshen Zhai

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A weak subsystem of Peano Arithmetic

Let $L_{\text{PA}} := \{0, +, \cdot, S, \leq\}$ be the first-order language of PA, whose axioms are the ‘basic axioms’ BASIC together with an axiom scheme IND, consisting of the induction axioms

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The L_{PA} -theory $I\Delta_0$ is the subtheory of PA axiomatized by BASIC and the restriction of IND to Δ_0 -formulas.

Polynomial Translation for $I\Delta_0$

Theorem (Paris-Wilkie, 1985; Krajíček, 1995)

Let R be a binary relation symbol and let $\varphi(x)$ be $\Delta_0(R)$ -formula. If $I\Delta_0(R) \vdash \forall x \varphi(x)$, then there is $d \in \mathbb{N}$ such that $S_{F_d}(\langle \varphi \rangle_n) = \text{poly}(n)$.

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Theorem (Cook, 1975)

Let $\varphi(x)$ be a Σ_1^b -formula. If $S_2^1 \vdash \forall x \varphi(x)$, then $S_{EF}(\|\varphi\|^n) = \text{poly}(n)$.

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- $I\Delta_0 \leftrightarrow$ Functions in the Linear Time Hierarchy LTH.

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We call φ a *defining formula* for f , and say that f is *provably total in* T if it is Σ_1 -definable function in T .

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Theorem (Buss's Witnessing Theorem, 1986)

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- 1. f is Σ_1^b -definable in S_2^1 , i.e., f is provably total in S_2^1 .*
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- 3. f is polynomial-time computable.*

The Engine of $I\Delta_0$: Parikh's Theorem

Theorem (Parikh, 1971)

Let $\varphi(\bar{x}, y)$ be a bounded formula. If $I\Delta_0 \vdash \forall \bar{x} \exists y \varphi(\bar{x}, y)$, then there is an L_{PA} -term $t(\bar{x})$ such that $I\Delta_0 \vdash \forall \bar{x} \exists y \leq t(\bar{x}) \varphi(\bar{x}, y)$.

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Suppose towards a contradiction that $I\Delta_0 \vdash \forall \bar{x} \exists y \varphi(\bar{x}, y)$ but $I\Delta_0 \not\vdash \forall \bar{x} \exists y \leq t(\bar{x}) \varphi(\bar{x}, y)$ for any term $t(\bar{x})$.

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Suppose towards a contradiction that $I\Delta_0 \vdash \forall \bar{x} \exists y \varphi(\bar{x}, y)$ but $I\Delta_0 \not\vdash \forall \bar{x} \exists y \leq t(\bar{x}) \varphi(\bar{x}, y)$ for any term $t(\bar{x})$. Then

$$I\Delta_0 \not\vdash \bigvee_i \forall \bar{x} \exists y \leq t_i(\bar{x}) \varphi(\bar{x}, y),$$

so the theory $I\Delta_0 \cup \{\forall y \leq t(\bar{c}) \neg \varphi(\bar{c}, y) : t \text{ term}\}$ is satisfiable by the Compactness Theorem, and thus has a model M . Define

$$N := \{a \in M : M \models a \leq t(\bar{c}), \text{ for some term } t\},$$

which models $I\Delta_0$.

The Engine of $I\Delta_0$: Parikh's Theorem

Theorem (Parikh, 1971)

Let $\varphi(\bar{x}, y)$ be a bounded formula. If $I\Delta_0 \vdash \forall \bar{x} \exists y \varphi(\bar{x}, y)$, then there is an L_{PA} -term $t(\bar{x})$ such that $I\Delta_0 \vdash \forall \bar{x} \exists y \leq t(\bar{x}) \varphi(\bar{x}, y)$.

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which models $I\Delta_0$. But $N \models \exists \bar{x} \forall y \neg \varphi(\bar{x}, y)$, contradiction. ■

Proof of the Witnessing Theorem for $I\Delta_0$, part 1

Theorem (Witnessing Theorem for $I\Delta_0$)

For any function $f : \mathbb{N}^k \rightarrow \mathbb{N}$, the following are equivalent.

- 1. f is Σ_1 -definable in $I\Delta_0$, i.e., f is provably total in $I\Delta_0$.*
- 2. f is Δ_0 -definable in $I\Delta_0$.*
- 3. There is a Δ_0 -defining formula $\varphi(\bar{x}, y)$ for f and a term $t(\bar{x})$ such that $I\Delta_0 \vdash \forall \bar{x} \exists! y \leq t(\bar{x}) \varphi(\bar{x}, y)$.*
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The Linear Time Hierarchy

Definition

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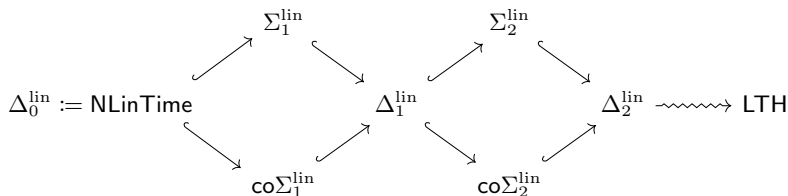
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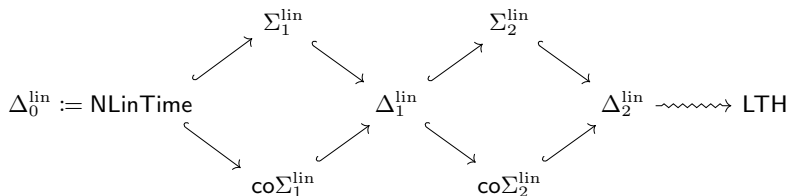


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Proof of the Witnessing Theorem for $I\Delta_0$, part 2

Theorem ($3 \Leftrightarrow 4$)

For any $f : \mathbb{N}^k \rightarrow \mathbb{N}$, there is a Δ_0 -defining formula $\varphi(\bar{x}, y)$ for f and a term $t(\bar{x})$ such that $I\Delta_0 \vdash \forall \bar{x} \exists! y \leq t(\bar{x}) \varphi(\bar{x}, y)$ iff $f \in \text{FLTH}$.

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Thank you!

- Buss S. R. (1986), *Bounded Arithmetic*, Naples, Bibliopolis. (Revision of the 1985 Princeton University Ph.D. thesis.)
- Buss S. R. (1999), *Bounded Arithmetic, Proof Complexity, and Two Papers of Parikh*, Annals of Pure and Applied Logic, **96**, pp. 43-55.
- Cook S. (1975), *Feasibly Constructive Proofs and the Propositional Calculus*, in: Proceedings of the 7th Annual ACM Symposium on Theory of Computing, pp. 83-97, ACM Press.
- Cook S. and Nguyen P. (2010), *Logical Foundations of Proof Complexity*, Perspectives in Logic, Cambridge University Press.
- Hájek P. and Pudlák P. (1993), *Metamathematics of First-order Arithmetic*, Perspectives in Mathematical Logic, Springer-Verlag.
- Krajíček J. (1995), *Bounded Arithmetic, Propositional Logic, and Complexity Theory*, Encyclopedia of Mathematics and its Applications, Cambridge University Press.
- Krajíček J. (1995), *On Frege and Extended Frege proof systems*, in: Feasible Mathematics II (Birkhäuser, Boston 1995), eds. P. Clote and J. Remmel, pp. 284-319.
- Parikh R. (1971), *Existence and Feasibility in Arithmetic*, J. Symbolic Logic, **36**, pp. 494-508.
- Paris J. and Wilkie A. (1985), *Counting Problems in Bounded Arithmetic*, in: Methods in Mathematical Logic (Caracas, Venezuela 1983), eds. A. Dold and B. Eckmann, LNM, vol. 1130, pp. 317-340, Springer-Verlag.
- Wrathall, C. (1978), *Rudimentary Predicates and Relative Computation*, SIAM J. Computing, **7**, pp. 194-209.